

Pack Indexing for Template Names

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Motivation

We added the ability to index packs of types and expressions in C++26 through [P2662R3](#) [3]. ([P2662R3](#) [3] is now implemented in Clang and GCC, and we got very positive feedback).

However, [P2662R3](#) [3] does not allow the indexing of a pack of templates. There is no good reason for that. The intent was always to be able to index all packs.

Both [P2841R7](#) [2] and [P2989R2](#) [1] were in flight, and it was not clear to me if either these papers would impact the indexing of packs of *template-names*. So, I punt that question to the present paper. It turns out that [P2841R7](#) [2] has no impact on the design of this paper - except that indexing a pack of concept template parameter just works - and [P2989R2](#) [1] was not approved for C++26.

In short, we are proposing to complete the design of pack indexing.

Design

The syntax for indexing a pack of template-name is similar to the syntax to the syntax used to index a pack of types or expressions.

```
template < template <typename> typename... TT>
struct S {
    using First = TT...[0];
};
```

The indexed pack is a *template-name* and can be used anywhere any *template-name* would be usable. All packs of template template parameters can be indexed (type, variable, concepts).

Implementation

This paper has not been implemented, but I am confident this can be implemented in Clang without trouble. I can't comment on other implementations.

Wording



Names of template specializations

[temp.names]

A template specialization[temp.spec] can be referred to by a *template-id*:

```
simple-template-id:
    template-name < template-argument-listopt >

template-id:
    simple-template-id
    operator-function-id < template-argument-listopt >
    literal-operator-id < template-argument-listopt >

template-name:
    identifier
    simple-template-name
    pack-index-template-name

pack-index-template-name:
    simple-template-name ... [ constant-expression ]

simple-template-name:
    identifier

template-argument-list:
    template-argument ...opt
    template-argument-list , template-argument ...opt

template-argument:
    constant-expression
    type-id
    nested-name-specifieropt template-name
    nested-name-specifier template template-name
```

The component name of a *simple-template-id*, *template-id*, or *template-name* is the first name in it.

The *simple-template-name* *P* in a *pack-index-template-name* shall denote a pack.

The *constant-expression* shall be a converted constant expression [expr.const] of type `std::size_t` whose value *V*, termed the index, is such that $0 \leq V < \text{sizeof} \dots (P)$.

A *pack-index-template-name* is a pack expansion [temp.variadic].

[Note: The *pack-index-template-name* denotes the type of the V^{th} element of the pack. — end note]

A < is interpreted as the delimiter of a *template-argument-list* if it follows a name that is not a *conversion-function-id* and

- that follows the keyword `template` or a `~` after a *nested-name-specifier* or in a class member access expression, or

- for which name lookup finds the injected-class-name of a class template or finds any declaration of a template, or
- that is an unqualified name for which name lookup either finds one or more functions or finds nothing, or
- that is a terminal name in a *using-declarator* [namespace.udecl], in a *declarator-id* [dcl.meaning], or in a type-only context other than a *nested-name-specifier* [temp.res].



Variadic templates

[temp.variadic]

In a template parameter pack that is a pack expansion [temp.param]:

- In a `sizeof... expression[expr.sizeof]`; the pattern is an *identifier*.
- In a *pack-index-expression*; the pattern is an *identifier*.
- In a *pack-index-specifier*; the pattern is a *typedef-name*.
- In a *pack-index-template-name*; the pattern is a *simple-template-name*.
- In a *fold-expression* [expr.prim.fold]; the pattern is the *cast-expression* that contains an unexpanded pack.
- In a fold expanded constraint[temp.constr.fold]; the pattern is the constraint of that fold expanded constraint.

[Editor's note: [...]]

The instantiation of a pack expansion considers items $E_1, E_2, \dots, E_N$, where N is the number of elements in the pack expansion parameters. Each E_i is generated by instantiating the pattern and replacing each pack expansion parameter with its i^{th} element. Such an element, in the context of the instantiation, is interpreted as follows:

- if the pack is a template parameter pack, the element is
 - a *typedef-name* for a type template parameter pack,
 - an *id-expression* for a constant template parameter pack, or
 - a *template-name* for a template template parameter pack
 designating the i^{th} corresponding type, constant, or template template argument;
- if the pack is a function parameter pack, the element is an *id-expression* designating the i^{th} function parameter that resulted from instantiation of the function parameter pack declaration;
- if the pack is an *init-capture* pack, the element is an *id-expression* designating the variable introduced by the i^{th} *init-capture* that resulted from instantiation of the *init-capture* pack declaration; otherwise
- if the pack is a structured binding pack, the element is an *id-expression* designating the i^{th} structured binding in the pack that resulted from the structured binding declaration.

When N is zero, the instantiation of a pack expansion does not alter the syntactic interpretation of the enclosing construct, even in cases where omitting the pack expansion entirely would otherwise be ill-formed or would result in an ambiguity in the grammar.

The instantiation of a `sizeof... expression[expr.sizeof]` produces an integral constant with value N .

When instantiating a *pack-index-expression* P , let K be the index of P . The instantiation of P is the *id-expression* E_K .

When instantiating a *pack-index-specifier* P , let K be the index of P . The instantiation of P is the *typedef-name* E_K .

When instantiating a *pack-index-template-name* P , let K be the index of P . The instantiation of P is the *simple-template-name* E_K .

[Editor's note: [...]]

◆ Type equivalence

[temp.type]

Two *template-ids* are the same if

- their *template-names*, *operator-function-ids*, or *literal-operator-ids* refer to the same template, and
- their corresponding type *template-arguments* are the same type, and
- the template parameter values determined by their corresponding constant template arguments[temp.arg.nontype] are template-argument-equivalent (see below), and
- their corresponding template *template-arguments* refer to the same template.

Two *template-ids* that are the same refer to the same class, function, or variable.

[Editor's note: [...]]

If an expression e is type-dependent [temp.dep.expr], `decltype( $e$ )` denotes a unique dependent type. Two such *decltype-specifiers* refer to the same type only if their *expressions* are equivalent[temp.over.link]. [Note: However, such a type might be aliased, e.g., by a *typedef-name*. — end note]

For a type template parameter pack $T, T, \dots$ [*constant-expression*] denotes a unique dependent type.

If the *constant-expression* of a *pack-index-specifier* is value-dependent, two such *pack-index-specifier*s refer to the same type only if their *constant-expressions* are equivalent [temp.over.link]. Otherwise, two such *pack-index-specifier*s refer to the same type only if their indexes have the same value.

If the *constant-expression* of a *pack-index-template-name* is value-dependent, two such *pack-index-template-names* refer to the same template only if their *constant-expressions* are equiv-

alent [temp.over.link]. Otherwise, two such *pack-index-template-names* refer to the same template only if their indexes have the same value.



Keywords

[gram.key]

New context-dependent keywords are introduced into a program by `typedef`[dcl.typedef], `namespace`[namespace.def], `class`[class], `enumeration`[dcl.enum], and `template`[temp] declarations.

typedef-name:
 identifier
 simple-template-id

namespace-name:
 identifier
 namespace-alias

namespace-alias:
 identifier

class-name:
 identifier
 simple-template-id

enum-name:
 identifier

template-name:
 identifier
 simple-template-name
 pack-index-template-name

Feature test macros

[*Editor's note: Bump `__cpp_pack_indexing` to the date of adoption*] .

- [1] Corentin Jabot and Gašper Ažman. P2989R2: A simple approach to universal template parameters. <https://wg21.link/p2989r2>, 6 2024.
 - [2] Corentin Jabot, Gašper Ažman, James Touton, and Hubert Tong. P2841R7: Concept and variable-template template-parameters. <https://wg21.link/p2841r7>, 2 2025.
 - [3] Corentin Jabot and Pablo Halpern. P2662R3: Pack indexing. <https://wg21.link/p2662r3>, 12 2023.
- [N5008] Thomas Köppe *Working Draft, Standard for Programming Language C++*
<https://wg21.link/N5008>